Introductory Econometrics

Multivariate Regression Model

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Sampling Distributions of the OLS Estimators

- **Classical linear model (CLM) assumptions**

$\underbrace{MLR.1 - MLR.5}$ "Gauss-Markov assumptions"

$\underbrace{MLR.1 - MLR.6}$ "Classical linear model (CLM) assumptions"

- u_i are independent and identically distributed (iid) as $Normal(0, \sigma^2)$

$$\hat{\beta}_j = \beta_j + \sum_{i=1}^{n} w_{ij} u_i \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

The estimators are normally distributed around the true parameters with the variance that was derived earlier

where

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

Testing Hypotheses About a Single Population Parameter

- **Theorem 4.1 (Normal Sampling Distributions)**

  Under the CLM Assumptions (and conditional on the sample outcomes of the explanatory variables),

$$\hat{\beta}_j \sim Normal[\beta_j, Var(\hat{\beta}_j)]$$

  and so

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim Normal(0,1)$$

  The standardized estimators follow a standard normal distribution

  to test hypotheses about β_j: $sd(\hat{\beta}_j)$ depends on $\sigma = sd(u)$, which is unknown. But we have $\hat{\sigma}$ as an estimator of σ.

Testing Hypotheses About a Single Population Parameter

- **Null hypothesis**

$$H_0 : \beta_j = 0$$

The population parameter is equal to zero, i.e. after controlling for the other independent variables, there is no effect of x_j on y

- **t-statistic (or t-ratio)**

The t-statistic will be used to test the null hypothesis.

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

The farther the estimated coefficient is away from zero, the less likely it is that the null hypothesis holds true. But what does "far" away from zero mean?

The t-statistic measures how many estimated standard deviations the estimated coefficient is away from zero.

- **Distribution of the t-statistic <u>if the null hypothesis is true</u>**

$$t_{\hat{\beta}_j} = \hat{\beta}_j / se(\hat{\beta}_j) = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$$

Testing Hypotheses About a Single Population Parameter

- **Goal: Define a rejection rule so that, if it is true, H_0 is rejected only with a small probability (= significance level, e.g. 5%)**

- **Testing against one-sided alternatives (greater than zero)**

  Traditional approach to hypothesis testing:

  - Choose a null hypothesis: $H_0 : \beta_j = 0$
  - Choose an alternative hypothesis: $H_1 : \beta_j > 0$
  - Choose a **significance level** (or simply **level**, or **size**) for the test. That is, the probability of rejecting the null hypothesis when it is in fact true (**Type I Error**) .
  - Calculate t-statistics $t_{\hat{\beta}_j}$
  - Choose a critical value, $c > 0$, so that the **rejection rule** is

$$t_{\hat{\beta}_j} > c$$

Testing Hypotheses About a Single Population Parameter

- **Testing more general hypotheses about a regression coefficient**
- **Null hypothesis**

$$H_0: \ \beta_j = a_j$$

Hypothesized value of the coefficient

- **t-statistic**

$$t = \frac{(estimate - hypothesized\ value)}{standard\ error} = \frac{(\hat{\beta}_j - a_j)}{se(\hat{\beta}_j)}$$

- **The test works exactly as before, except that the hypothesized value is substracted from the estimate when forming the statistic**

Testing Hypotheses About a Single Population Parameter

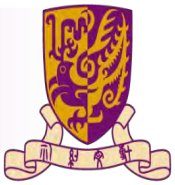
- **"Statistically significant" variables in a regression**
  - If a regression coefficient is different from zero in a two-sided test, the corresponding variable is said to be "statistically significant"
  - If the number of degrees of freedom is large enough so that the normal approximation applies, the following rules of thumb apply:

$|t - stat.| > 1.645 \longrightarrow$ "statistically significant at 10 % level"

$|t - stat.| > 1.96 \longrightarrow$ "statistically significant at 5 % level"

$|t - stat.| > 2.576 \longrightarrow$ "statistically significant at 1 % level"

Testing Hypotheses About a Single Population Parameter



- **Computing p-values for t-tests**
  - If the significance level is made smaller and smaller, there will be a point where the null hypothesis cannot be rejected anymore
  - The smallest significance level at which the null hypothesis is still rejected, is called the p-value of the hypothesis test
  - **p-value in two-sided test,**

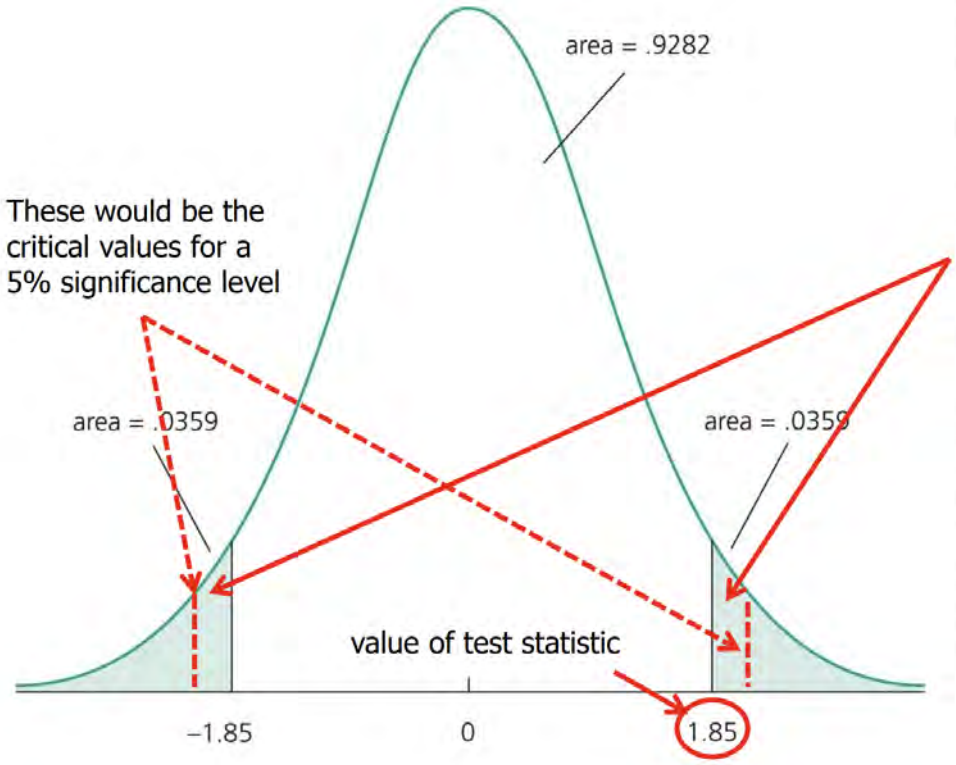
$$p\text{-value} = P(|T| > |t|)$$

where t is the value of the $t - statistic$ and T is a random variable with the t_{n-k-1} distribution.

  - Perhaps its most useful interpretation is the following: the p-value is the probability of observing a $t - statistic$ as extreme as we did if the null hypothesis is true

Testing Hypotheses About a Single Population Parameter

- **How the p-value is computed (here: two-sided test)**



The p-value is the smallest significance level at which the null hypothesis is still rejected.

In the two-sided case, the p-value is thus the probability that the t-distributed variable takes on a larger absolute value than the realized value of the test statistic, e.g.:

$$P(|T| > 1.85) = 2(.0.359) = .0718$$

From this, it is clear that a null hypothesis is rejected if and only if the corresponding p-value is smaller than the significance level.

For example, for a significance level of 5% the t-statistic would not lie in the rejection region.

Testing Hypotheses About a Single Population Parameter

- **Computing p-values for t-tests (cont.)**
  - A small p-value is evidence against the null hypothesis because one would reject the null hypothesis even at small significance levels
  - A large p-value is evidence in favor of the null hypothesis
  - P-values are more informative than tests at fixed significance levels
  - Given p-value, we can carry out a test at any significance level. If α is the chosen level, then

$$\text{Reject } H_0 \text{ if } p\text{-value} < \alpha$$



Testing Hypotheses About a Single Population Parameter

- **Computing p-values for one-sides alternatives**

$$\text{one-sided } p\text{-value} = \frac{\text{two-sided } p\text{-value}}{2}$$

- **Example: Factors Affecting NBA Salaries**

Data: NBASAL.DTA

Annual salary, thousands $

Averge games per year

Averge minutes per game

Averge points per game

$$lwage = \beta_0 + \beta_1 games + \beta_2 avgmin + \beta_3 points + \beta_4 rebounds + \beta_5 assists + u$$

Averge rebounds per game

Averge assists per game

Testing Hypotheses About a Single Population Parameter

- **Example: Factors Affecting NBA Salaries (cont.)**

```
. reg lwage games avgmin points rebounds assists
```

Source	SS	df	MS
Model	90.2698185	5	18.0539637
Residual	117.918945	263	.448361006
Total	208.188763	268	.776823743

Number of obs	= 269
F(5, 263)	= 40.27
Prob > F	= 0.0000
R-squared	= 0.4336
Adj R-squared	= 0.4228
Root MSE	= .6696

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
games	.0004132	.002682	0.15	0.878	-.0048679	.0056942
avgmin	.0302278	.0130868	2.31	0.022	.0044597	.055996
points	.0363734	.0150945	2.41	0.017	.0066519	.0660949
rebounds	.0406795	.0229455	1.77	0.077	-.0045007	.0858597
assists	.0003665	.0314393	0.01	0.991	-.0615382	.0622712
_cons	5.648996	.1559075	36.23	0.000	5.34201	5.955982

Confidence Intervals

- **Confidence intervals**
- Using the fact that $(\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$, simple manipulation leads to a 95% confidence interval for the unknown β_j

Critical value of two-sided test

$$P\left(\hat{\beta}_j - c_{0.05} \cdot se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + c_{0.05} \cdot se(\hat{\beta}_j)\right) = 0.95$$

Lower bound of the Confidence interval

Upper bound of the Confidence interval

Confidence level

- **Interpretation of the confidence interval**
  - The bounds of the interval are random
  - In repeated samples, the interval that is constructed in the above way will cover the population regression coefficient in 95% of the cases

Confidence Intervals

- **Confidence intervals for typical confidence levels**

$$P\left(\hat{\beta}_j - c_{0.01} \cdot se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + c_{0.01} \cdot se(\hat{\beta}_j)\right) = 0.99$$

$$P\left(\hat{\beta}_j - c_{0.05} \cdot se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + c_{0.05} \cdot se(\hat{\beta}_j)\right) = 0.95$$

$$P\left(\hat{\beta}_j - c_{0.10} \cdot se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + c_{0.10} \cdot se(\hat{\beta}_j)\right) = 0.90$$

Use rules of thumb $c_{0.01} = 2.576, c_{0.05} = 1.96, c_{0.10} = 1.645$

- **Relationship between confidence intervals and hypotheses tests**

$a_j \notin interval \Rightarrow$ reject $H_0 : \beta_j = a_j$ in favor of $H_1 : \beta_j \neq a_j$

$a_j \in interval \Rightarrow$ fail to reject $H_0 : \beta_j = a_j$

significance level $= 100 -$ confidence level

Confidence Intervals

- **Example: Factors Affecting NBA Salaries**

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Testing Single Linear Restrictions

- **Testing hypotheses about a linear combination of parameters**
- **Example: Return to education at 2 year college vs. at 4 year university**

Data: TWOYEAR.dta



Years of education at 2 year colleges

Years of education at 4 year university

$$\log(wage) = \beta_0 + \beta_1 jc + \beta_2 univ + \beta_3 exper + u$$

Test $H_0 : \beta_1 - \beta_2 = 0$ against $H_1 : \beta_1 - \beta_2 < 0$.

A possible test statistic would be:

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{se(\hat{\beta}_1 - \hat{\beta}_2)}$$

The difference between the estimates is normalized by the estimated standard deviation of the difference. The null hypothesis would have to be rejected if the statistic is "too negative" to believe that the true difference between the parameters is equal to zero.

Testing Single Linear Restrictions

- **Difficult to compute with standard regression output**

$$se(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{\widehat{Var}(\hat{\beta}_1 - \hat{\beta}_2)}$$

$$= \sqrt{\widehat{Var}(\hat{\beta}_1) + \widehat{Var}(\hat{\beta}_2) - 2\widehat{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$$

Usually not available in regression output, need to be estimated separately in Stata

```
. vce

Covariance matrix of coefficients of regress model
```

e(V)	jc	univ	exper	_cons
jc	.00004663			
univ	1.928e-06	5.330e-06		
exper	-1.718e-08	3.933e-08	2.480e-08	
_cons	-.00001741	-.00001573	-3.105e-06	.00044353

One can calculate $se(\hat{\beta}_1 - \hat{\beta}_2) = 0.0069357$

Testing Single Linear Restrictions

- **Alternative method**

Define $\theta_1 = \beta_1 - \beta_2$ and test $H_0 : \theta_1 = 0$ against $H_1 : \theta_1 < 0$.

Insert into original regression

$$\log(wage) = \beta_0 + (\theta_1 + \beta_2)jc + \beta_2 univ + \beta_3 exper + u$$

$$= \beta_0 + \theta_1 jc + \beta_2 (jc + univ) + \beta_3 exper + u$$

a new regressor (= total years of college and university)

$$se(\hat{\theta}_1) = se(\hat{\beta}_1 - \hat{\beta}_2)$$

Testing Single Linear Restrictions

- **Estimation results**

Total years of college and university

$$\widehat{\log(wage)} = \underset{(.021)}{1.472} \underset{(.0069)}{- .0102}\, jc + \underset{(.0023)}{.0769}\, totcoll + \underset{(.0002)}{.0049}\, exper$$

$$n = 6,763, \quad R^2 = .222$$

$$t = -.0102/.0069 = -1.48$$

Hypothesis is rejected at 10% level but not at 5% level

$$p - value = P(t - ratio < -1.48) = .070$$

$$-.0102 \pm 1.96(.0069) = (-.0237, .0003)$$

- This method works always for single linear hypotheses
- **Easist approach in Stata is to use commend:** ***lincom***

Testing Multiple Exclusion Restrictions

- **Testing multiple linear restrictions: The F-test**
- **Testing exclusion restrictions**

Example: Factors Affecting MBL Salaries (MBL1.dta)



Salary of major league base ball player

Years in the league

Average number of games per year

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr$$

$$+\beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

Batting average

Home runs per year

Runs batted in per year

$H_0 : \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$ against $H_1 : H_0$ is not true

Test whether performance measures have no effect/can be exluded from regression.

Testing Multiple Exclusion Restrictions

- A test of multiple restrictions is called a **multiple hypotheses test** or a **joint hypotheses test.**
- The original model, containing all variables, is the **unrestricted model**:

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

- When we impose $H_0 : \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$, we get the **restricted model:**

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + u$$

Testing Multiple Exclusion Restrictions

- **Estimation of the unrestricted model**

$$\widehat{\log}(salary) = \underset{(0.29)}{11.19} + \underset{(.0121)}{.0689}\, years + \underset{(.0026)}{.0126}\, gamesyr$$

$$+ \underset{(.00110)}{.00098}\, bavg + \underset{(.0161)}{.0144}\, hrunsyr + \underset{(.0072)}{.0108}\, rbisyr$$

None of these variabels is statistically significant when tested individually

$$n = 353,\ SSR = 183.186,\ R^2 = .6278$$

Idea: How would the model fit be if these variables were dropped from the regression?

A hint: Multicollinearity ($hrunsyr$ and $rbisyr$)

Testing Multiple Exclusion Restrictions

- **Estimation of the restricted model**

$$\widehat{\log}(salary) = \underset{(0.11)}{11.22} + \underset{(.0125)}{.0713}\, years + \underset{(.0013)}{.0202}\, gamesyr$$

$$n = 353,\ SSR = 198.311,\ R^2 = .5971$$

The sum of squared residuals necessarily increases, but is the increase statistically significant?

- **Test statistic**

Numerator degrees of freedom $= df_r - df_{ur}$

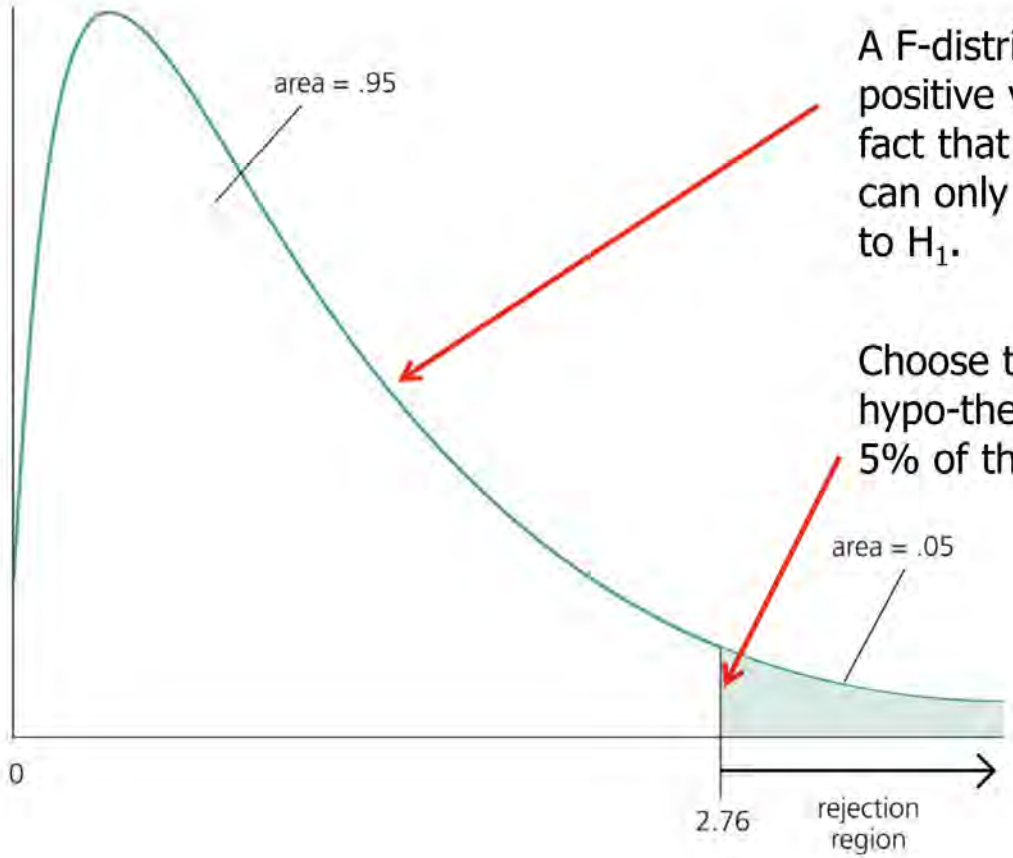
$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} \sim F_{q,n-k-1}$$

The relative increase of the sum of squared residuals when going from H_0 to H_1 follows a F-distribution (if the null hypothesis H_0 is correct)

Denominator degrees of freedom $= df_{ur}$

Testing Multiple Exclusion Restrictions

- The 5% critical value and rejection region in an $F_{3,60}$ distribution



A F-distributed variable only takes on positive values. This corresponds to the fact that the sum of squared residuals can only increase if one moves from H_0 to H_1.

Choose the critical value so that the null hypo-thesis is rejected in, for example, 5% of the cases, although it is true.

Testing Multiple Exclusion Restrictions

- **Test decision in example**

$$F = \frac{(198.311 - 183.186)/3}{183.186/(353 - 5 - 1)} \approx 9.55$$

Number of restrictions to be tested

Degrees of freedom in the unrestricted model

$$F \sim F_{3,347} \Rightarrow c_{0.01} = 3.78$$

$$P(F - statistic > 9.55) = 0.000$$

The null hypothesis is overwhelmingly rejected (even at very small significance levels).

- **Discussion**
  - The three variables are "jointly significant"
  - They were not significant when tested individually
  - The likely reason is multicollinearity between them

Testing Multiple Exclusion Restrictions

- **Relationship between F and t statistics**

  When we apply the F statistic to the case of testing significance of a single independent variable, i.e., $q = 1$

$$F_{1,n-k-1} = t^2_{n-k-1}$$

- **R-squared form of the F Statistic**

  The same dependent variable is used in the two regression,

$$\begin{aligned} SSR_r &= (1 - R_r^2)SST \\ SSR_{ur} &= (1 - R_{ur}^2)SST \end{aligned}$$

  therefore,

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$$

Testing Multiple Exclusion Restrictions

- **Example: Parents' education in a birth weight equation**

  Data: BWGHT.dta

Birth weight, in pounds

Number of cigarettes the mother smoked per day

Birth order of this child

Annual family income

$$bwght = \beta_0 + \beta_1 cigs + \beta_2 parity + \beta_3 faminc + \beta_4 motheduc + \beta_5 fatheduc + u$$

Years of schooling for the mother

Years of schooling for the father

$H_0 : \beta_4 = 0, \beta_5 = 0$ against $H_1 : H_0$ is not true

Test whether parents' education has no effect on birth weight.

Testing Multiple Exclusion Restrictions

- **Test decision in example**

$$R^2_{ur} = 0.0387 \qquad R^2_r = 0.0364$$

$$n = 1191$$

Number of restrictions to be tested

$$F = \frac{(0.0387 - 0.0364)/2}{(1 - 0.0387)/1185} = 1.44$$

Degrees of freedom in the unrestricted model

$$F \sim F_{2,1185} \Rightarrow c_{0.05} = 3$$

$$P(F - statistic > 1.44) = 0.238$$

The null hypothesis cannot be rejected

Testing Multiple Exclusion Restrictions

- **Test of overall significance of a regression**

$$y = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + u$$

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$$

The null hypothesis states that the explanatory variables are not useful at all in explaining the dependent variable

$$y = \beta_0 + u$$

Restricted model (regression on constant)

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,n-k-1}$$

- **The test of overall significance is reported in Stata; the null hypothesis is usually overwhelmingly rejected**

Testing Multiple Exclusion Restrictions

- **Testing general linear restrictions with the F-test**
- **Example: Test whether house price assessments are rational**

Actual house price

The assessed housing value (before the house was sold)

Size of lot (in feet)

$$\log(price) = \beta_0 + \beta_1 \log(assess) + \beta_2 \log(lotsize)$$

$$+ \beta_3 \log(sqrft) + \beta_4 bdrms + u$$

Square footage

Number of bedrooms



$$H_0 : \beta_1 = 1, \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$$

If house price assessments are rational, a 1% change in the assessment should be associated with a 1% change in price.

In addition, other known factors should not influence the price once the assessed value has been controlled for.

Testing Multiple Exclusion Restrictions

- **Unrestricted regression**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_3 x_3 + \beta_4 x_4 + u$$

- **Restricted regression**

The restricted model is actually a regression of [y-x_1] on a constant

$$y = \beta_0 + x_1 + u \;\Rightarrow\; [y - x_1] = \beta_0 + u$$

- **Test statistic**

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{(1.880 - 1.822)/4}{1.822/(88-4-1)} \approx .661$$

$$F \sim F_{4,83} \;\Rightarrow\; c_{0.05} = 2.50 \;\Rightarrow\; H_0 \text{ cannot be rejected}$$

Testing Multiple Exclusion Restrictions

- **Regression output for the unrestricted regression**

$$\widehat{\log(price)} = \underset{(.570)}{.264} + \underset{(.151)}{1.043}\ \log(assess) + \underset{(.0386)}{.0074}\ \log(lotsize)$$

$$- \underset{(.1032)}{.1384}\ \log(sqrft) + \underset{(.0221)}{.0338}\ bdrms$$

$$n = 88, SSR = 1.822, R^2 = .773$$

When tested individually, there is also no evidence against the rationality of house price assessments

- **The F-test works for general multiple linear hypotheses**
- **For all tests and confidence intervals, validity of assumptions MLR.1 – MLR.6 has been assumed. Tests may be invalid otherwise.**

Testing Multiple Exclusion Restrictions

- **Reporting regression results** (Factors Affecting MLB Salaries)

	Dependent Variable: log(salary)			
VARIABLES	(1)	(2)	(3)	(4)
years	0.069***	0.068***	0.070***	0.071***
	(5.684)	(5.592)	(5.776)	(5.703)
gamesyr	0.013***	0.016***	0.011***	0.020***
	(4.742)	(10.079)	(5.202)	(15.023)
bavg	0.001	0.001	0.001	---
	(0.887)	(1.331)	(0.691)	
hrunsyr	0.014	0.036***	---	---
	(0.899)	(4.964)		
rbisyr	0.011	---	0.017***	---
	(1.500)		(5.117)	
Constant	11.192***	11.021***	11.275***	11.224***
	(38.752)	(41.476)	(41.197)	(103.625)
Observations	353	353	353	353
R-squared	0.628	0.625	0.627	0.597

Testing Multiple Exclusion Restrictions

- **Conclusion:**
  - Sampling Distributions of the OLS Estimators
  - Testing Hypotheses About a Single Population Parameter
  - Confidence Intervals
  - Testing Single Linear Restrictions
  - Testing Multiple Exclusion Restrictions